

HING ONG

Curriculum Vitae
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Updated on September 6, 2026
<https://hingong.github.io/>

EDUCATION

PhD	University at Albany, State University of NY, Atmospheric Sciences Dissertation: “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics”	2020
MS	National Taiwan University, Atmospheric Sciences Thesis: “Effects of artificial local compensation of convective mass flux in the cumulus parameterization”	2016
BS	National Taiwan University, Atmospheric Sciences	2014

HIGHLIGHTS

Research Experience

- Authored scientific papers
- Developed atmospheric models
- Coordinated a cross-institutional research project

Teaching & Training Experience

- Trained AI models on reasoning of Physics
- Taught undergraduate and graduate courses

PUBLICATIONS

Peer-Reviewed Publications in Atmospheric Sciences

- 2026 **Ong, H.**, Larson, V. E., The Coriolis effects in parameterization of atmospheric turbulence and convection. *J. Adv. Model. Earth Syst.*, Accepted.
- 2026 Zhang, A., Yang, D., **Ong, H.**, & Tan, Z., Understanding the evolution of global atmospheric rivers with a vapor kinetic energy framework. *Geophys. Res. Lett.*, 53, e2025GL119841.
- 2025 **Ong, H.**, Scale analysis for the Madden–Julian oscillation. *Q. J. R. Meteorol. Soc.*, 151(772), e5028.
- 2025 **Ong, H.**, & Yang, D., Westward- or eastward-propagating Rossby waves: Schematic illustrations. *J. Atmos. Sci.*, 82(9), 1903–1910.

- 2024 **Ong, H.**, & Yang, D, Vapor kinetic energy for the detection and understanding of atmospheric rivers. *Nat. Commun.*, 15, 9428.
- 2022 **Ong, H.**, & Yang, D., The compressional beta effect and convective system propagation. *J. Atmos. Sci.*, 79(8), 2031–2040.
- 2021 Skamarock, W. C., **Ong, H.**, & Klemp, J. B., A fully compressible nonhydrostatic deep-atmosphere equations solver for MPAS. *Mon. Weather Rev.*, 149(2), 571–583.
- 2020 **Ong, H.**, Comments on “On the structure and formation of UTLS PV dipole/jetlets in tropical cyclones by convective momentum surges”. *Mon. Weather Rev.*, 148(11), 4693–4695.
- 2020 **Ong, H.**, & Roundy, P. E., The compressional beta effect: Analytical solution, numerical benchmark, and data analysis. *J. Atmos. Sci.*, 77(11), 3721–3732.
- 2020 **Ong, H.**, & Roundy, P. E., Nontraditional hypsometric equation. *Q. J. R. Meteorol. Soc.*, 146(727), 700–706.
- 2019 **Ong, H.**, & Roundy, P. E., Linear effects of nontraditional Coriolis terms on intertropical convergence zone forced large-scale flow. *Q. J. R. Meteorol. Soc.*, 145(723), 2445–2453.
- 2017 **Ong, H.**, Wu, C. M., & Kuo, H. C., Effects of artificial local compensation of convective mass flux in the cumulus parameterization. *J. Adv. Model. Earth Syst.*, 9(4), 1811–1827.

In-Progress Works in Atmospheric Sciences

- 2026 **Ong, H.**, The nontraditional Coriolis terms and trade-wind cumuli. Preprint. doi:10.22541/essoar.176945330.02654752/v1
- 2026 **Ong, H.**, Jung, C., Wang, J., Kotamarthi, V. R., & Sever, G., Evaluation of near-surface air temperature and surface energy fluxes in a convection-permitting dynamical downscaling over the contiguous United States. Submitted.
- 2026 **Ong, H.**, Hughes, O., Herrington, A., Jablonowski, C., Lauritzen, P. H., & Yang, D., ITCZ and the nontraditional Coriolis terms. Abstract.

Peer-Reviewed Publication in Linguistics

- 2025 **Ong, H.**, Functional aspiration in Taiwanese. *Taiwan Journal of Linguistics*, 23(2), 51–81.

HONORS AND AWARDS

- 2020 **Climate and Global Change Postdoctoral Fellowship**, NOAA (declined)

RESEARCH EXPERIENCE

Independent Scholar	2025 to current
Coordinated a cross-institutional atmospheric model intercomparison project	
Brought CESM developers at NCAR into the project	
Brought E3SM developers at Univ. of Michigan into the project	
Brought CLUBB developers at Univ. of Wisconsin-Milwaukee into the project	
Framed scientific questions for the project	
Convened regular meetings for the project	
Formulated a minimal model for the Madden–Julian oscillation	
Formulated the Coriolis effects in atmospheric turbulence parameterization	
Designed a model hierarchy for deep-atmosphere dynamical cores	
Designed a model hierarchy for turbulence parameterization with the Coriolis effects	
Adapted the dynamics of CLUBB atmospheric turbulence parameterization	
Adapted the model interface of CESM, MPAS-A, and CLUBB	
Conducted sensitivity simulations of Coriolis effects with CESM with MPAS-A	
Conducted sensitivity simulations of Coriolis effects with SAM	
Conducted sensitivity simulations of Coriolis effects with CLUBB	
Guided ChatGPT to review source codes of MPAS atmospheric model	
Postdoctoral Appointee , Argonne National Laboratory	2023 to 2025
Performed climate dynamical downscaling with WRF model	
Evaluated the surface fluxes of the model against observations	
Conducted sensitivity simulations of land models in WRF	
Managed petabytes of data storage	
Postdoctoral Scholar , University of California, Davis	2020 to 2023
Adapted the dynamics of SAM atmospheric model.	
Performed spectral analysis to model simulation data.	
Formulated the prognostic equation of vapor kinetic energy.	
Analyzed MERRA2 and ERA5 reanalysis data.	
Student Researcher , University at Albany, State University of NY	2017 to 2020
Formulated a numerical idealized circulation model.	
Analyzed rawinsonde and ERA-Interim reanalysis data.	
Derived analytical equatorial wave solutions.	
Developed a benchmarking test for model dynamics.	
Adapted the dynamics of MPAS atmospheric model.	
Research Assistant , National Taiwan University	2016 to 2017
Participated in a scientific planning group in a field experiment.	
Composed a progress report.	
Student Researcher , National Taiwan University	2014 to 2016

Formulated a cumulus parameterization scheme.
Adapted the dynamics and physics of WRF atmospheric model.

TEACHING & TRAINING EXPERIENCE

AI Trainer , Handshake AI Created graduate Physics problems for AI to fail on reasoning Provided correct answers and reasonings for AI training	2026 to current
Adjunct Faculty , University of Maryland Global Campus Fundamentals of Physics I Fundamentals of Physics II	2026 to current
STEM Teacher , Seattle Taiwanese Language Association Designed STEM activities for K-to-11 students in Taiwanese Taigi Delivered STEM activities at Taiwanese Language Summer Camps	2023 to current
Teaching Assistant , University at Albany, State University of NY Applications of Subseasonal to Seasonal Dynamics Ocean Science Water and Climate Change Atmospheric Dynamics	2018 to 2020
Teaching Assistant , National Taiwan University Lab. of Synoptic Meteorology (<i>de facto</i> Lecturer) Fluid Mechanics Program and Scientific Computing	2014 to 2016

PROFESSIONAL SERVICE

Peer-Reviewed Articles for:

Journal of Atmospheric Sciences
Geophysical Research Letters
Monthly Weather Review
Journal of Geophysical Research: Atmospheres
Journal of Climate

Coordinated Seminar Series for:

2022 Winter Atmospheric Science Seminar, University of California, Davis

Panel-Reviewed STEM Education Materials for:

教育部國家教育研究院「臺灣台語新詞、專有名詞及學術名詞對譯審譯組」
(Taiwanese Taigi New Words, Proper Nouns, and Academic Nouns Translation Panel,
National Academy for Educational Research, Ministry of Education, Taiwan)

INVITED LECTURES (SELECTED)

- 2025 “Revisiting tropical dynamics: New insights on the MJO and double-ITCZ bias”
Physical Sciences Lab., NOAA, Boulder, CO, Jun 12.
Dept. of Atmospheric and Climate Science, Univ. of Washington, Seattle, WA, Jun 23
- 2025 “ITCZ and the nontraditional Coriolis terms”
CESM Workshop, Boulder, CO, Jun 9.
Climate & Global Dynamics Lab., NCAR, Boulder, CO, Feb 5.
- 2024 “Pressure perturbation in mesoscale meteorology”
Dept. of Geography and Meteorology, Valparaíso Univ., Valparaíso, IN, Mar 25.
- 2022 “Káng 風 soat 雨 òe 大氣” (Talk about wind, rain, and atmosphere)
Sè-kài Tâi-oân Bûn-hòa Lûn-tôaⁿ (World Taiwanese Culture Forum), Online, Nov 12.
Delivered in Taiwanese Taigi.
- 2021 “The nontraditional Coriolis terms and convective system propagation”
Geophysical Fluid Dynamics Lab., NOAA, Princeton, NJ, Sep 23.
- 2020 “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics”
Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan, Jan 10.
Dept. of Atmospheric Sciences, National Taiwan Univ., Taipei, Taiwan, Jan 9.
- 2019 “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics,”
Dept. of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA, Oct 30.
Mesoscale and Microscale Meteorology Lab., NCAR, Boulder, CO, Jul 25.

LANGUAGES

English: Professionally proficient

Taiwanese Taigi: Native (my official name since Dec 2021, Hing Ong)

Chinese Mandarin: Native (my official name until Dec 2021, Heng Wang)

OUTSTANDING SKILLS

Model Formulation: using partial differential equations.

Model Development: using Fortran, Matlab, or Python

Data Analysis: using Fortran, Matlab, NCL, Python, or Grads

RESEARCH INTERESTS

Geophysical Fluid Dynamics

Earth System Modeling